```
annanya@annanya-VirtualBox:~/OS-Jackfruit/boilerplate$ cd ~/OS-Jackfruit/boil
rplate
sudo insmod monitor.ko
ls -l /dev/container_monitor
[sudo] password for annanya:
insmod: ERROR: could not insert module monitor.ko: File exists
crw------- 1 root root 240, 0 Apr 16 11:59 /dev/containannanya@annanya-Virtu
```

1)**Module Loaded** `ls -l /dev/container_monitor` confirming the kernel module loaded and device was created

2) both containers listed with their metadeta.**Multi-Container Supervision** `sudo ./engine ps` showing alpha and beta containers tracked with PID, state, and memory limits.

```
[7] 7219
Supervisor started. rootfs=./rootfs socket=/tmp/mini_runtime.sock
Started container alpha pid=7227
Started container beta pid=7232
ID               PID      STATE      EXIT     SOFT_MB      HARD_MB
---------------- -------- ---------- -------- ------------ ------------
beta             7232     exited     0        40           64
alpha            7227     exited     0
```

3)logs output

Bounded-Buffer Logging `sudo ./engine logs alpha/beta` showing container output captured through the logging pipeline.

```
annanya@annanya-VirtualBox:~/OS-Jackfruit/boilerplate$
sudo ./engine logs alpha
Log: logs/alpha.log
hello from alpha
hello from alpha
annanya@annanya-VirtualBox:~/OS-Jackfruit/boilerplate$
sudo ./engine logs beta
Log: logs/beta.log
hello from beta
hello from beta
annanya@annanya-VirtualBox:~/OS-Jackfruit/boilerplate$
```

4)**CLI and IPC** `sudo ./engine start` command sent to supervisor over UNIX socket, supervisor responds with PID.

```
annanya@annanya-VirtualBox:~/OS-Jackfruit/boilerplate$ sudo ./engine start demo ./rootfs "
echo hello from demo"
Started container demo pid=8601
annanya@annanya-VirtualBox:~/OS-Jackfruit/boilerplate$
```

5 and 6)soft and hard kill

CLI and IPC `sudo ./engine start` command sent to supervisor over UNIX socket, supervisor responds with PID.

```
annanya@annanya-VirtualBox:~/OS-Jackfruit/boilerplate$ sudo dmesg | grep container_monitor | tail -20
[  455.306696] [container_monitor] Module loaded. Device: /dev/container_monitor
[ 2332.579839] [container_monitor] Registering container=alpha pid=6488 soft=41943040 hard=67108864
[ 2338.998571] [container_monitor] Registering container=beta pid=6497 soft=41943040 hard=67108864
[ 2526.980046] [container_monitor] Unregister request container=alpha pid=6488
[ 2527.029380] [container_monitor] Unregister request container=beta pid=6497
[ 4048.772064] [container_monitor] Registering container=alpha pid=7227 soft=41943040 hard=67108864
[ 4048.877271] [container_monitor] Registering container=beta pid=7232 soft=41943040 hard=67108864
[ 4155.812144] [container_monitor] Registering container=alpha pid=7263 soft=41943040 hard=67108864
[ 4156.442254] [container_monitor] Registering container
```

7) **Scheduling Experiment** `top` showing highpri (nice=-10) receiving ~2-3x more CPU than lowpri (nice=+10).

```
top -b -n 1 | grep cpu_hog
cp: cannot stat 'cpu_hog': No such file or directory
Started container highpri pid=7680
Started container lowpri pid=7687
ID                  PID      STATE      EXIT     SOFT_MB      HARD_MB
---------------- -------- ---------- -------- ------------ ------------
lowpri              7687     exited     127      40           64
```

```
Started container lowpri pid=7700
annanya@annanya-VirtualBox:~/OS-Jackfruit/boilerplate$ sudo ./engine ps
top -b -n 1 | grep cpu_hog
ID                  PID      STATE      EXIT     SOFT_MB      HARD_MB
---------------- -------- ---------- -------- ------------ ------------
lowpri              7700     exited     127      40           64
lowpri              7687     exited     12
```

8)no defunct process and container stopped.**Clean Teardown** `ps aux | grep defunct` showing zero zombie processes after all containers and supervisor exited.

```
ackfruit/boilerplate$ s stop highpri
udo ./engine stop highpri
sudo ./engine stop lowprit2
sudo ./engine stop memtest2
sudo ./engine psfunct
ps aux | grep defunct
Container highpri not running
Container lowpri not running
Container memtest2 not running
ID                  PID      STATE      EXIT     SOFT_MB      HARD_MB
---------------- -------- ---------- -------- ------------ ------------
memtest2            7744     exited     127      5            10
lowpri              7700     exited     12
annanya     7777  0.0  0.1   9080  2636 pts/2    R+   18:40   0:00 grep --color=auto defunct
```